



Loan Approval Prediction – Project Report

1. Introduction

Loan approval is a critical process in banking and financial institutions. The goal of this project is to build a machine learning model that predicts whether a loan application should be approved or rejected based on applicant financial and demographic information.

This system helps reduce manual effort, improves decision consistency, and minimizes financial risk for lending institutions.

2. Problem Statement

Given historical loan application data, develop a classification model that predicts:

- **1 → Loan Approved**
- **0 → Loan Rejected**

The model should maximize accuracy while ensuring high precision and recall for risk-sensitive financial decisions.

3. Dataset Description

Dataset: Loan Approval Prediction Case Study

Source: Kaggle

Link: <https://www.kaggle.com/datasets/bhanupratapbiswas/loan-approval-prediction-case-study>

Features:

- Applicant Income
- Loan Amount
- Loan Term
- CIBIL Score
- Residential Asset Value
- Commercial Asset Value
- Luxury Asset Value
- Bank Asset Value

- Education
 - Employment Status
 - Loan Status (Target Variable)
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4. Methodology

4.1 Data Preprocessing

- Handled missing values using:
 - Median (numerical features)
 - Mode (categorical features)
 - Encoded categorical variables using Label Encoding / One-Hot Encoding
 - Applied feature scaling using StandardScaler
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4.2 Exploratory Data Analysis (EDA)

Key insights:

- Loan approval depends strongly on **CIBIL score**
- Higher income applicants are more likely to get approved
- Applicants with higher asset value show better approval rates
- Dataset showed slight class imbalance

Visualizations used:

- Loan approval distribution
 - Income vs approval
 - CIBIL score distribution
 - Correlation heatmap
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4.3 Handling Class Imbalance

To improve model performance:

- SMOTE (Synthetic Minority Oversampling Technique) was applied
- Alternative approaches tested:

- Class weights
- Undersampling

SMOTE gave the most balanced performance.

5. Machine Learning Models

Three classification models were trained and compared:

5.1 Logistic Regression

A baseline linear model used for interpretability and quick benchmarking.

5.2 Decision Tree Classifier

Captures non-linear patterns but prone to overfitting.

5.3 Random Forest Classifier

An ensemble model combining multiple decision trees for improved accuracy and stability.

6. Model Evaluation

Evaluation metrics used:

- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

7. Results

Model	Precision	Recall	F1 Score	ROC-AUC
Logistic Regression	0.82	0.80	0.81	0.87
Decision Tree	0.79	0.78	0.78	0.83
Random Forest	0.90	0.88	0.89	0.94

8. Best Model Selection

✓ **Selected Model: Random Forest Classifier**

Reason:

- Highest ROC-AUC (0.94)
- Best balance between precision and recall
- Stable performance on unseen data
- Handles non-linear relationships effectively

9. Feature Importance

Most influential features:

1. CIBIL Score
2. Annual Income
3. Loan Amount
4. Asset Values
5. Employment Status

Insight:

Credit score and financial strength are the strongest predictors of loan approval.

10. Business Insights

- Applicants with high CIBIL scores have significantly higher approval probability.
- Loan amount relative to income is a key risk indicator.
- Asset ownership improves approval chances.
- Model can reduce manual loan processing time significantly.

11. Deployment Strategy

A probability-based decision threshold is recommended:

- **0.50** → Balanced decision-making
- **0.60 – 0.70** → Low-risk lending (recommended for banks)
- **0.40** → Higher approval rate but increased risk

For financial institutions, a threshold around **0.65** is ideal to minimize defaults.

12. Conclusion

This project successfully demonstrates an end-to-end machine learning pipeline for loan approval prediction. Among all tested models, Random Forest performed best and is recommended for deployment.

The system can assist financial institutions in making faster, data-driven, and more reliable lending decisions.

13. Technologies Used

- Python
 - Pandas, NumPy
 - Matplotlib, Seaborn
 - Scikit-learn
 - Imbalanced-learn (SMOTE)
 - Jupyter Notebook / Google Colab
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14. Future Improvements

- Hyperparameter tuning using GridSearchCV
 - Try advanced models like XGBoost / LightGBM
 - Deploy model using Flask / Streamlit
 - Build real-time loan approval web app
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